

ID No: _____ Name: _____ Sec No.: _____

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

II SEMESTER 2025-26

CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing

25-02-2026 (Wednesday)

QUIZ-3 (closed book)

B

Max time: 15 Min

Max Marks: 08

Q1. For an 8086 processor, write a four-line program to set the Auxiliary carry flag (AF) of the flag register. Use one PUSH, OR and a POP instruction. AF is bit number '4' of the flag register (Note: Least significant bit is bit '0'). Other bits of Flag register should not be modified [2M]

```
PUSHF
MOV BP, SP
OR WORD PTR [BP], 0010H
POPF
```

1M if pushf /popf is written,

1m for rest

Q2. For an 8086 processor assume the following, BX=2000H. The memory contents are

Data segment	Data
DS:2000H	34H
DS:2001H	12H
DS:2002H	78H
DS: 2003H	56H

The following instruction is executed: JMP DWORD PTR [BX]. What will be the value of IP after the execution of the instruction?[1m]

1234H

Q3. For an 8086 processor, answer the questions given in the table for the snippet of the program given in next page.

Question	Answer in hex
Value of SP before executing MOV AX, AAAAH instruction [2M]	<u>FFF4</u>
Value of SP before executing POP DX instruction [1M]	<u>FFFE</u>
Final Value of CX after executing the program (TILL JMP L1) [1M]	<u>2222</u>
Final Value of DX after executing the program (TILL JMP L1) [1M]	<u>GIVE 1 MARK</u> <u>IF above 3</u> <u>answers are</u> <u>correct</u>

Assume all segment registers are initialized to 0859h. SP is pointing to FFEh

```
.model tiny
.DATA
SAVE_REGS MACRO
    PUSH AX
    PUSH DX
ENDM
.code
.startup
    MOV AX, 1111H
    MOV BX, 2222H
    MOV CX, 3333H
    MOV DX, 4444H
    PUSH BX
    PUSH CX
    POP AX
    CALL SUB5
    POP DX
L1: JMP L1
.EXIT
SUB5 PROC NEAR USES BX
    SAVE_REGS
    MOV AX, AAAAH
    MOV BX, BBBBH
    MOV DX, CCCCH
    POP AX
    POP BX
    POP CX
    RET
```

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A

Max time: 15 Min

Max Marks: 08

Q1. For an 8086 processor, write a four-line program to set the Parity flag (PF) of the flag register. Use one PUSH, OR and a POP instruction. PF is bit number '2' of the flag register (Note: Least significant bit is bit '0'). Other bits of Flag register should not be modified [2M]

```
PUSHF
MOV BP, SP
OR WORD PTR [BP], 0004H
POPF
```

1M if pushf /popf is written,

1m for rest

Q2. For an 8086 processor assume the following, BX=2000H. The memory contents are

Data segment	Data
DS:2000H	34H
DS:2001H	12H
DS:2002H	78H
DS: 2003H	56H

The following instruction is executed: JMP DWORD PTR [BX]. What will be the value of CS after the execution of the instruction? [1m]

5678H

Q3. For an 8086 processor, answer the questions given in the table for the snippet of the program given in next page.

Question	Answer in hex
Value of SP before executing MOV DX, CCCCH instruction [2M]	10F4H
Value of SP after executing POP DX instruction [1M]	1100H
Final Value of DX after executing the program (TILL JMP L1) [1M]	<u>GIVE 1 MARK</u> <u>IF other 3</u> <u>answers are</u> <u>correct</u>
Final Value of AX after executing the program(TILL JMP L1) [1M]	4444H

Assume all segment registers are initialized to 0859h. SP is pointing to 10FEh

```
.model tiny
.DATA
SAVE_REGS MACRO
    PUSH AX
    PUSH DX
ENDM
.code
.startup
    MOV AX, 1111H
    MOV BX, 2222H
    MOV CX, 3333H
    MOV DX, 4444H
    PUSH BX
    PUSH CX
    POP AX
    CALL SUB5
    POP DX
L1: JMP L1
.EXIT
SUB5 PROC NEAR USES BX
    SAVE_REGS
    MOV AX, AAAAH
    MOV BX, BBBBH
    MOV DX, CCCCH
    POP AX
    POP BX
    POP CX
    RET
SUB5 ENDP
END
```

